

Confidence interval on the mean, ^{with} variance ~~are~~ unknown.

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2 , then $100(1-\alpha)\%$

Confidence Interval on μ is given by,

$$\bar{x} - t_{(\alpha/2, n-1)} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{(\alpha/2, n-1)} \frac{s}{\sqrt{n}}$$

where $t_{(\alpha/2, n-1)}$ is the upper $100\alpha/2\%$ of the t distribution with $(n-1)$ degrees of freedom.

2: The load at specimen failure is as follows:
19.8, 15.4, 11.4, 19.5, 10.1, 18.5, 14.1, 8.8, 14.9, 7.9,
17.6, 13.6, 7.5, 12.7, 16.7, 11.9, 15.4, 11.9, 15.8,
11.4, 15.4, 11.4 are the sample values of a normal distribution. Find 95% confidence interval of μ .

$$\text{Given } 100(1-\alpha)\% = 95\%$$

$$1-\alpha = 0.95$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$\text{Sample size, } n = 22$$

$$t_{(\alpha/2, n-1)} = t_{(0.025, 21)}$$

$$\text{For two-tailed test, } t = 2.414$$

$$\text{one-tailed test, } t = 2.080$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{301.7}{22} = \underline{\underline{13.71}}$$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
19.8	6.09	37.088
15.4	1.69	2.856
11.4	-2.31	5.33
19.5	5.79	33.52
10.1	-3.61	13.03
18.5	4.79	22.94
14.1	0.39	0.152
8.8	-4.91	24.10
14.9	1.19	1.416
7.9	-5.81	33.75
17.6	3.89	15.13
13.6	-0.11	0.012
7.5	-6.21	38.56
12.7	-1.01	1.02
16.7	2.99	8.94
11.9	-1.81	3.27
15.4	1.69	2.85
11.9	-1.81	3.27
15.8	2.09	4.36
11.4	-2.31	5.33
15.4	1.69	2.856
11.4	-2.31	5.33
301.7		

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$\sum (x_i - \bar{x})^2 = 265.11$$

$$s^2 = \frac{265.11}{22} = 12.05$$

$$s^2 = 12.05$$

$$\therefore s = \sqrt{12.05} = \underline{\underline{3.471}}$$

Confidence Interval on μ is

$$\bar{x} - t_{(\alpha/2, n-1)} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{(\alpha/2, n-1)} \frac{s}{\sqrt{n}}$$

For two-tailed test,

$$13.71 - 2.414 \times \frac{3.47}{\sqrt{22}} \leq \mu \leq 13.71 + 2.414 \times \frac{3.47}{\sqrt{22}}$$

$$13.71 - (2.414 \times 0.739) \leq \mu \leq 13.71 + (2.414 \times 0.739)$$

$$\underline{\underline{11.924 \leq \mu \leq 15.495}}$$

For one-tailed test,

$$13.71 - 2.080 \times \frac{3.47}{\sqrt{22}} \leq \mu \leq 13.71 + 2.080 \times \frac{3.47}{\sqrt{22}}$$

$$\underline{\underline{12.17 \leq \mu \leq 15.248}}$$